

HP's built-in leap motion

HP's latest PC with a Leap Motion sensor allows users to perform tasks without touching it <http://dgit.in/14uyyh9>

Shape-shifting lens

A fluid-filled lens that can adjust focus like a human eye is in the works at Ohio State University

HOW WE TESTED

Android tablets have been around for quite some time now and its only in the past year that we started seeing devices that are priced around ₹20,000 and boast of a quad-core processor giving you a smooth experience along with the ability to run all the apps available on the Google Play store. A tablet today is essentially used for a few primary functions including web browsing, watching movies and videos, playing games, running utility apps, the occasional spreadsheet, document typing/editing, with some social networking thrown into the mix. Gaming on the tablet too has become quite addictive especially since graphically heavy games such as Real Racing 3 and Dead Trigger have found their way onto people's fingertips. To ensure that a tablet can perform these tasks well, we used a series of subjective and objective tests to measure their performance.

Benchmarks

The objective tests are a series of benchmarks that measure the performance of the hardware powering the tablets. These benchmarks are Quadrant, Antutu, 3DMark, GFXBench and SmartBench. They give us an objective idea of how the tablets will perform when used for productivity and multitasking. The benchmarks also give us an understanding of how games will perform on the tablet, especially the ones that demand a lot of power under the hood. There is also a test that measures the efficiency of the preloaded browser on the device called the Octane Browser Benchmark. This test essentially tells us how the browser will perform especially if you are one who likes to browse heavy-duty websites.

The above-mentioned benchmarks give out a score (higher is better) and objectively measure the performance of the hardware.

Gaming

One of the biggest uses of a tablet today is to play games on the go. Some of the games that are available for tablets are graphically heavy and need some power to run smoothly. As a standard, we played quite a bit of Dead Trigger, Angry Birds Star Wars, Real Racing 3 and Temple Run 2. These are some of the most popular games on the Google Play store and represent elements from the graphically heavy (Dead Trigger and Real Racing 3) to the average everyday games (Temple Run 2). While playing these games, we look out for issues such as screen lag, input lag, stutter etc. There are also benchmarks, 3D Mark and GFX Bench (Egypt HD on screen and off screen) that measures the gaming capabilities of the tablet. In all these cases, a higher score means better performance.

Display

There are two ways in which we test the display of the tablet. The subjective test is where we interact with the display, play games, type, browse the web, read an eBook, and more to see how responsive the display is. As an objective test, we use an app

called Display Tester Pro. With the help of the app, we measure the colour banding, contrast and saturation to see if the reproduction of content on the display is accurate. The test also helps us measure the viewing angles, white noise and grey noise produced on the display of the tablet.

Start up

Beyond benchmarks, we also tested the start-up time of these devices to see how long it takes for the tablets to reach the lock screen from the power off mode. This was done after factory resetting the tablets, doing the required logins at the first start-up, and restarting the device again. No apps were installed prior to this test, except for the ones that came preloaded.

Multimedia

Since today's tablets offer multimedia capabilities, we ran a bunch of videos on the tablet. We tested the devices with all the popular formats (MP4, MKV and AVI). The videos included SD as well as HD content. We judged the ability of the device to play the video formats smoothly – jumping forwards and backward on the play-timeline to check for stutters, freeze, crashes etc. The quality and experience offered by the display was taken into account as well – viewing angles, colour reproduction, crispness, brightness etc. We also played a bunch of MP3 songs to see how the speakers performed.

Calling capabilities

Some of the tablets available in this price range have the ability to make calls. Although this isn't a primary function of the tablet, its presence is valued nonetheless. To see how the tablet performed with its calling capabilities, we used the device as a phone in a variety of situations such as noisy environments, absolute silence and while commuting. Here we take into account the crispness of the call quality and the signal strength.

Build

Since a tablet is considered as a "lifestyle" device, we have also taken into consideration the build quality, plastics used, slimness of the device and its overall feel in our hands. There are multiple parameters on which each tablet is rated, before we reach one final build quality score.

Battery life

To test the longevity of the battery on the tablet, we ran a video in loop to see how long it lasts. The battery was charged to 100%. We ran an MP4 720p movie on loop with the brightness and audio at maximum. The amount of time taken by the device to reach 0% battery level is determined as the battery life. Please note that if you use the device more conservatively (switch off location services, reduce the brightness and volume, etc.), it will last longer.